

Shree Pandit

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EDUCATION

Maulana Azad National Institute of Technology, Bhopal	Expected May 2028
B.Tech - Computer Science and Engineering	CGPA: 7.40
The Himalaya Academy, Dewas	2023
Madhya Pradesh Board of Secondary Education – Class XII	Percentage: 85.2%
Carmel High School, Gadchiroli	2021
Central Board of Secondary Education – Class X	Percentage: 93.8%

WORK EXPERIENCE

Data Engineering Intern	May 2026 – July 2026
Transform Rural India Foundation (TRIF)	
- Centralized 186,000+ unstructured survey responses into 7 structured fact tables, resolving schema inconsistencies and encoding errors via a grain-aware star-schema ETL pipeline in Python (Pandas, NumPy). - Accelerated KPI reporting from weekly exports to near-real-time across 2,000+ Gram Panchayats in 5 states, by designing a multi-page Power BI dashboard tracking 150+ metrics. - Automated multi-source ingestion across public portals by engineering resilient scrapers with retry logic and rate-limiting via Python (Selenium), eliminating 15+ hours/week of manual entry.	

PROJECTS

A.R.G.U.S C++, Python, FastAPI, React, Tailwind, ZeroMQ, TimescaleDB, Redis	GitHub
AI-driven Recognition Gateway for Unidirectional IP Security in Critical Infrastructure Networks.	
- Engineered a C++ deep-packet inspection engine extracting JA3 fingerprints, DNS entropy, and host fan-out features on unidirectional links, achieving sub-5 ms per-flow feature extraction with zero packet loss under sustained 1 Gbps capture. - Architected a ZeroMQ PUB/SUB pipeline with a schema-translating consumer that streams DPI features into a live SOC dashboard at < 20 ms end-to-end latency, guaranteeing zero message loss under sustained 1 Gbps capture with automatic back-pressure handling. - Delivered alert broadcasting with automatic Redis-failover recovery, by building a FastAPI backend with Redis Pub/Sub and WebSocket fan-out, backed by a fail-closed PostgreSQL/TimescaleDB persistence layer.	
Auralis C++, Python, FastAPI, React.js, Npcap, WebSockets	GitHub
Real-time network traffic diagnostic instrument featuring cross-platform C++/Python IPC.	
- Achieved sub-10ms parsing of raw L2/L4 network frames, by engineering a C++ capture engine (libpcap/Npcap) streaming 5-tuple flow data via a custom TCP IPC bridge. - Built an async FastAPI aggregation layer backed by a fixed 300-second in-memory ring buffer that maintains live flow state and serves sub-10 ms metric queries with no disk I/O under concurrent load. - Rendered live bandwidth and connection states at 1-second intervals, by developing a React/Tailwind dashboard with an SVG oscilloscope trace and a WebSocket-driven patch-panel visualization.	
NITBFreshers Study Portal JavaScript, Tailwind CSS, PHP, MySQL	GitHub
Full-stack academic resource management system for MANIT students.	
- Sustained 1,000+ active users with zero downtime, deploying a modular backend and indexing relational queries to minimize read contention on shared infrastructure. - Cut document load times by 40%, replacing synchronous reloads with asynchronous Fetch API calls and integrating PDF.js for in-browser document streaming. - Secured 100% of academic assets against unauthorized access, enforcing session validation via 64-character HttpOnly tokens and parameterized database operations.	

RESEARCH EXPERIENCE

End-to-End Geotechnical Data Extraction Framework	August 2026 – Present
Guided by Dr. Kishan Dharavat, Dept. of Civil Engineering, MANIT Bhopal	
- Designed a hybrid EfficientNet-b0 + rule-based pipeline parsing 2,400+ unstructured geotechnical PDFs into tabular records, reaching 0.98 F1 and slashing extraction time from hours to under 90s per file. - Fine-tuned a Faster R-CNN backbone for dense document layout analysis, achieving 0.95 mAP on spatial bounding-box detection to resolve overlapping tabular structures.	

OPEN SOURCE CONTRIBUTIONS

EDMS – Endpoint Data Management System Rust, Axum, React.js	GitHub
Open-source project by HashedTokens – Core Contributor.	
- Synchronized decoupled React frontend and Rust backend, eliminating fabricated client-side state by binding the UI to live REST APIs and bidirectional WebSocket streams. - Restored end-to-end endpoint testing functionality, resolving critical IPC fallback failures by patching Axum dispatchers and migrating HTTP client routines to the active child process.	

TECHNICAL SKILLS

Languages: C++, Python, C, JavaScript, SQL
Systems & Frameworks: React.js, Node.js, FastAPI, ZeroMQ, WebSockets, libpcap/Npcap
ML & Data Libraries: PyTorch, Scikit-learn, Pandas, NumPy, Power BI
Databases & Storage: PostgreSQL, TimescaleDB, Redis, MySQL
Developer Tools: Git, Docker, Linux Shell
CS Coursework: Data Structures & Algorithms, Computer Networks, Operating Systems, DBMS, OOP, Data Science, Statistics

ACHIEVEMENTS

- Solved 1,500+ problems in 150+ contests, holding Guardian (LeetCode), Expert (Codeforces), and 4-Star (CodeChef) ranks.
- 2nd Runner-Up at Kinetic Coding Challenge by IIFE Visakhapatnam.
- National Finalist at Shaastra Programming Contest (IIT Madras) and Code Wars, Prometeo' 26 (IIT Jodhpur).

POSITION OF RESPONSIBILITY

Corporal, 12 MP BN NCC, MANIT Bhopal	August 2024 – Present
Team Management Head, ISTE SC MANIT	February 2025 – Present